

ENRICO TRANQUILLINI

Machine Learning Engineer | Training & Evaluation Systems

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 4+ years building production ML systems in C++/CUDA and PyTorch. Strong fit for Nous Research's open-source AI mission: turns models, datasets, evaluation loops, and deployment code into reliable infrastructure. Systems-level background across ONNX Runtime, real-time inference, edge hardware, and control engineering; international experience across Italy, China, and the US.

SKILLS

Machine Learning:	PyTorch, TensorFlow, Lightning, Scikit-learn, ONNX Runtime, model evaluation
Training & Data:	training pipelines, dataset curation, annotation review, synthetic workflows, MLflow
Infrastructure:	C++ inference services, CUDA acceleration, Docker, CI/CD, Azure, edge systems
Computer Vision:	OpenCV, YOLO, real-time inference, end-to-end CV systems
Programming:	Python, C++, CUDA, Rust, C#
Languages:	English (Fluent), Italian (Native), French (Basic)

EXPERIENCE

Machine Learning Engineer

Nov 2022 – Present

Danieli Automation | Buttrio, Italy

- **Training & Deployment Infrastructure:** Reduced training time for new deployments by **40%** by standardising the model-development path across **PyTorch** and **C++**, with reproducible handoff into ONNX Runtime production inference.
- **Low-Latency ML Systems:** Deployed end-to-end inference systems across **multiple machines and instances**, achieving sub-100ms latency per frame in C++/CUDA for real-time steel-plant quality control.
- **Evaluation & Data Tooling:** Built internal annotation, audit, and dataset-review tools that accelerated error analysis and shortened the loop between data quality, model iteration, and deployment readiness.

Visiting Researcher

Oct 2019 – Dec 2019

University of California, Irvine – Earth System Science Dept. | Irvine, CA, USA

- **Distributed-Systems Measurement:** Applied statistical tests on time-series data to pinpoint hardware bottlenecks in drone-ground-station communication, enabling a **17% reduction in multi-hop latency** (150ms → 125ms).

Junior Manufacturing Engineer (Intern)

Oct 2018 – Jan 2019

Jabil Circuit (Shanghai) Ltd. | Shanghai, China

- **QA & Risk Mitigation:** Supervised SMT production-line QA, detecting and resolving process defects in-cycle to maintain deliverable schedules and reduce risk of future production delays.

ACHIEVEMENTS & PROJECTS

Double Degree – Politong Project Sep 2016 – Jan 2019 | Milan, Italy & Shanghai, China

Politecnico di Milano & **Tongji University**

- Selected for the **Politong** joint programme — one of very few students chosen for this competitive double-degree track combining Italian and Chinese engineering curricula.
- Awarded double-degree recognition across both institutions.

JENIA AI Hackathon Top 3 of 20 teams

- Developed a **multi-agent AI workflow** using **n8n**, integrating tool use and automated orchestration for end-to-end task execution.

Open-Source

- Active on GitHub (ML systems, computer vision, control, MLOps) and Codeberg, maintaining C++ forks of the Darknet/YOLO stack: darknet, DarkHelp, DarkMark.

EDUCATION

Politecnico di Milano | Milan, Italy

B.Sc. – M.Sc. – Automation and Control Engineering

Mar 2015 – Dec 2023

Tongji University | Shanghai, China

B.Sc. – Computer Science and Engineering

Sep 2016 – Jan 2019

OPEN-SOURCE PROJECTS

Profiles: github.com/iltranqui | codeberg.org/iltranqui

darknet | C++ | <https://codeberg.org/iltranqui/darknet>

Darknet/YOLO object detection framework. Development branch adding **FP8 training**, **instance segmentation** and **oriented bounding box** support, plus **3D bounding-box** inference.

DarkHelp | C++ | <https://codeberg.org/iltranqui/DarkHelp>

C++ wrapper library for Darknet. Expanded with APIs for the new segmentation and oriented-bounding-box tasks, so they can be used from C++ applications without touching Darknet internals.

DarkMark | C++ | <https://codeberg.org/iltranqui/DarkMark>

Image annotation tool for building Darknet datasets. Updated to annotate **segmentation** masks, **oriented bounding boxes** and **3D bounding boxes**.